

# DOKUMENTACJA GAIA

## 1. Opis i Wymagania

Współczesne budynki mieszkalne o dużym zagęszczeniu użytkowników, takie jak akademiki, są szczególnie narażone na problem infestacji szkodników, w tym karaluchów. Zjawisko stanowi nie tylko problem higieniczny, ale również wpływa na komfort życia studentów oraz generuje koszty po stronie administracji budynku. Obecnie w akademikach stosowane metody zwalczania infestacji mają głównie charakter reaktywny — administracja podejmuje działania dopiero po otrzymaniu zgłoszenia od mieszkańców. Takie podejście prowadzi do opóźnień w eliminacji szkodników i sprzyja dalszemu rozprzestrzenianiu się kolonii.

- System musi być w stanie zidentyfikować obecność karaluchów w monitorowanym obszarze za pomocą czujników ruchu.
- Kiedy czujnik lub urządzenie agentowe ma obowiązek regularnie przesyłać dane do systemu centralnego lub agenta koordynującego.
- Przesyłane dane z czujników powinny obejmować informacje o liczbie wykrytych osobników, lokalizacji i czasie zdarzenia.
- System powinien analizować napływające dane, zidentyfikować wzorce pojawiania się karaluchów.
- System powinien wskazywać miejsca o największej aktywności, aby zoptymalizować działania likwidacyjne.
- W przypadku wykrycia aktywności karaluchów system powinien powiadamiać firmę dezynsekcyjną, poszkodowanych studentów i administrację.
- System powinien poinformować studentów o złym stanie w ich pokojach i potrzebie poprawienia stanu.

## 2. Model Ról

### Role: SensorNode

|                  |                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | A physical device equipped with motion sensors that detects cockroach presence in a specific area and periodically transmits data to the CentralCoordinator.                   |
| Responsibilities | <b>Liveness:</b> SensorNodeToCentralCoordinatorDataTransfer^ω<br><b>Safety:</b> UniqueID(SensorNodeID) != Null<br>CockroachCount >= 0<br>Location != Null<br>Timestamp != Null |
| Permissions      | READ:<br>WRITE: SensorNodeID, CockroachCount, Location, Timestamp<br>CONSUME:<br>PRODUCE: SensorNodeID, CockroachCount, Location, Timestamp                                    |
| Activities       |                                                                                                                                                                                |
| Protocols        | SensorNodeToCentralCoordinatorDataTransfer                                                                                                                                     |

### Role: CentralCoordinator

|                  |                                                                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | The central system component responsible for collecting, aggregating, and processing data from SensorNodes to identify high-activity zones and trigger notifications.                                                                                                                                     |
| Responsibilities | <b>Liveness:</b> (AggregateSensorData . CentralCoordinatorToPatternRecognitionModuleAnalysis . IdentifyHighActivityZones)^ω<br><b>Safety:</b> UniqueID(SensorNodeID) != Null<br>CockroachCount >= 0<br>Location != Null<br>Timestamp != Null<br>AggregatedSensorData != Null<br>HighActivityZones != Null |
| Permissions      | READ: SensorNodeID, CockroachCount, Location, Timestamp<br>WRITE: AggregatedSensorData, HighActivityZones<br>CONSUME: SensorNodeID, CockroachCount, Location, Timestamp<br>PRODUCE: AggregatedSensorData, HighActivityZones                                                                               |
| Activities       | AggregateSensorData, IdentifyHighActivityZones                                                                                                                                                                                                                                                            |
| Protocols        | CentralCoordinatorToPatternRecognitionModuleAnalysis                                                                                                                                                                                                                                                      |

### Role: PatternRecognitionModule

|                  |                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | An analytical module that processes sensor data to detect and analyze cockroach activity patterns and generates reports for the CentralCoordinator.             |
| Responsibilities | <b>Liveness:</b> (AnalyzeActivityPatterns . GenerateReports)^ω<br><b>Safety:</b> AggregatedSensorData != Null<br>Timestamp != Null<br>HighActivityZones != Null |
| Permissions      | READ: AggregatedSensorData, Timestamp<br>WRITE: HighActivityZones<br>CONSUME: AggregatedSensorData, Timestamp<br>PRODUCE: HighActivityZones                     |
| Activities       | AnalyzeActivityPatterns, GenerateReports                                                                                                                        |
| Protocols        |                                                                                                                                                                 |

### Role: NotificationService

|                  |                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | A service responsible for sending alerts to external entities such as pest control companies, students, and building administration based on detected activity. |
| Responsibilities | <b>Liveness:</b> (ComposeAlert . DispatchAlerts)^ω<br><b>Safety:</b> AlertDetails != Null<br>Location != Null<br>Timestamp != Null                              |
| Permissions      | READ: AlertDetails, Location, Timestamp<br>WRITE:<br>CONSUME: AlertDetails, Location, Timestamp<br>PRODUCE:                                                     |
| Activities       | ComposeAlert, DispatchAlerts                                                                                                                                    |
| Protocols        | NotificationServiceToPestControlCompanyAlert, NotificationServiceToBuildingAdministrationAlert, NotificationServiceToStudentResidentAlert                       |

### Role: BuildingAdministration

|                  |                                                                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | The administrative body responsible for managing the building and responding to pest control alerts.                                                |
| Responsibilities | <b>Liveness:</b> NotificationServiceToBuildingAdministrationAlert^ω<br><b>Safety:</b> AlertDetails != Null<br>Location != Null<br>Timestamp != Null |
| Permissions      | READ: AlertDetails, Location, Timestamp<br>WRITE:<br>CONSUME: AlertDetails, Location, Timestamp<br>PRODUCE:                                         |
| Activities       | RespondToAlert                                                                                                                                      |
| Protocols        | NotificationServiceToBuildingAdministrationAlert                                                                                                    |

### Role: PestControlCompany

|                  |                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | The external service provider tasked with eliminating cockroach infestations upon notification.                                                 |
| Responsibilities | <b>Liveness:</b> NotificationServiceToPestControlCompanyAlert^ω<br><b>Safety:</b> AlertDetails != Null<br>Location != Null<br>Timestamp != Null |
| Permissions      | READ: AlertDetails, Location, Timestamp<br>WRITE:<br>CONSUME: AlertDetails, Location, Timestamp<br>PRODUCE:                                     |
| Activities       | ExecuteTreatment                                                                                                                                |
| Protocols        | NotificationServiceToPestControlCompanyAlert                                                                                                    |

### Role: StudentResident

|                  |                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Description      | A resident of the building who may receive notifications about pest issues in their room or surrounding areas.                               |
| Responsibilities | <b>Liveness:</b> NotificationServiceToStudentResidentAlert^ω<br><b>Safety:</b> AlertDetails != Null<br>Location != Null<br>Timestamp != Null |
| Permissions      | READ: AlertDetails, Location, Timestamp<br>WRITE:<br>CONSUME: AlertDetails, Location, Timestamp<br>PRODUCE:                                  |
| Activities       | ReceiveNotification                                                                                                                          |
| Protocols        | NotificationServiceToStudentResidentAlert                                                                                                    |

### 3. Model Interakcji

| Global                                        | Name                                                 | Initiator           | Responder                | Inputs                                            | Outputs           | Description                                                                                                                                   |
|-----------------------------------------------|------------------------------------------------------|---------------------|--------------------------|---------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| SensorNode sends data to CentralCoordinator   | SensorNodeToCentralCoordinatorDataTransfer           | SensorNode          | CentralCoordinator       | SensorNodeID, CockroachCount, Location, Timestamp |                   | A SensorNode transmits its cockroach activity data to the CentralCoordinator for aggregation and analysis.                                    |
| CentralCoordinator activity                   | CentralCoordinatorToPatternRecognitionModuleAnalysis | CentralCoordinator  | PatternRecognitionModule | AggregatedSensorData, Timestamp                   | HighActivityZones | The CentralCoordinator forwards aggregated sensor data to the PatternRecognitionModule for recognition and high-activity zone identification. |
| NotificationService to PestControlCompany     | NotificationServiceToPestControlCompanyAlert         | NotificationService | PestControlCompany       | AlertDetails, Location, Timestamp                 |                   | The NotificationService dispatches an alert to the PestControlCompany when cockroach activity is detected in high-risk zones.                 |
| NotificationService to BuildingAdministration | NotificationServiceToBuildingAdministrationAlert     | NotificationService | BuildingAdministration   | AlertDetails, Location, Timestamp                 |                   | The NotificationService informs BuildingAdministration about cockroach activity for administrative response.                                  |
| NotificationService to StudentResident        | NotificationServiceToStudentResidentAlert            | NotificationService | StudentResident          | AlertDetails, Location, Timestamp                 |                   | The NotificationService notifies StudentResident about pest activity in their room or nearby area.                                            |

## 4. Model Usage

| Service                               | Agent                         | In/Out                                                                                                      | Pre/Post Condition                                                                                                                                                                  | Derived From                                         |
|---------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| SendSensorData                        | SensorNodeAgent               | <b>IN:</b> SensorNodeID, CockroachCount, Location, Timestamp<br><b>OUT:</b>                                 | <b>PRE:</b> SensorNodeID != Null AND CockroachCount >= 0 AND Location != Null AND Timestamp != Null<br><b>POST:</b> Data successfully transmitted to CentralCoordinator             | SensorNodeToCentralCoordinatorDataTransfer           |
| AggregateSensorData                   | CentralCoordinatorAgent       | <b>IN:</b> SensorNodeID, CockroachCount, Location, Timestamp<br><b>OUT:</b> AggregatedSensorData, Timestamp | <b>PRE:</b> SensorNodeID != Null AND CockroachCount >= 0 AND Location != Null AND Timestamp != Null AND UniqueID(SensorNodeID) != Null<br><b>POST:</b> AggregatedSensorData != Null | AggregateSensorData                                  |
| ForwardDataToPatternRecognitionModule | CentralCoordinatorAgent       | <b>IN:</b> AggregatedSensorData, Timestamp<br><b>OUT:</b> HighActivityZones                                 | <b>PRE:</b> AggregatedSensorData != Null AND Timestamp != Null<br><b>POST:</b> HighActivityZones != Null                                                                            | CentralCoordinatorToPatternRecognitionModuleAnalysis |
| AnalyzeActivityPatterns               | PatternRecognitionModuleAgent | <b>IN:</b> AggregatedSensorData, Timestamp<br><b>OUT:</b> HighActivityZones                                 | <b>PRE:</b> AggregatedSensorData != Null AND Timestamp != Null<br><b>POST:</b> HighActivityZones != Null                                                                            | AnalyzeActivityPatterns                              |
| GenerateReports                       | PatternRecognitionModuleAgent | <b>IN:</b> AggregatedSensorData, Timestamp<br><b>OUT:</b> HighActivityZones                                 | <b>PRE:</b> AggregatedSensorData != Null AND Timestamp != Null<br><b>POST:</b> HighActivityZones != Null                                                                            | GenerateReports                                      |
| IdentifyHighActivityZones             | CentralCoordinatorAgent       | <b>IN:</b> AggregatedSensorData, Timestamp<br><b>OUT:</b> HighActivityZones                                 | <b>PRE:</b> AggregatedSensorData != Null AND Timestamp != Null AND UniqueID(SensorNodeID) != Null<br><b>POST:</b> HighActivityZones != Null                                         | IdentifyHighActivityZones                            |
| ComposeAlert                          | NotificationServiceAgent      | <b>IN:</b> AlertDetails, Location, Timestamp<br><b>OUT:</b>                                                 | <b>PRE:</b> AlertDetails != Null AND Location != Null AND Timestamp != Null<br><b>POST:</b> Alert composed and ready for dispatch                                                   | ComposeAlert                                         |
| DispatchAlerts                        | NotificationServiceAgent      | <b>IN:</b> AlertDetails, Location, Timestamp<br><b>OUT:</b>                                                 | <b>PRE:</b> AlertDetails != Null AND Location != Null AND Timestamp != Null<br><b>POST:</b> Alerts dispatched to all relevant parties                                               | DispatchAlerts                                       |
| ExecuteTreatment                      | PestControlCompanyAgent       | <b>IN:</b> AlertDetails, Location, Timestamp<br><b>OUT:</b>                                                 | <b>PRE:</b> AlertDetails != Null AND Location != Null AND Timestamp != Null<br><b>POST:</b> Treatment executed at specified location                                                | ExecuteTreatment                                     |
| RespondToAlert                        | BuildingAdministrationAgent   | <b>IN:</b> AlertDetails, Location, Timestamp<br><b>OUT:</b>                                                 | <b>PRE:</b> AlertDetails != Null AND Location != Null AND Timestamp != Null<br><b>POST:</b> Administrative response initiated                                                       | RespondToAlert                                       |
| ReceiveNotification                   | StudentResidentAgent          | <b>IN:</b> AlertDetails, Location, Timestamp<br><b>OUT:</b>                                                 | <b>PRE:</b> AlertDetails != Null AND Location != Null AND Timestamp != Null<br><b>POST:</b> Student notified about pest issue in their area                                         | ReceiveNotification                                  |

## 5. Model Znajomości

